Fiscal Year:	FY 2007	**Task Last Updated:**	FY 04/09/2007
PI Name:	Cannell, David S Ph.D.		
Project Title:	GRADIENT DRIVEN FLUCTUATIONS		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Santa Barbara	**State:**	CA
Zip Code:	93106-9530	**Congressional District:**	23
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2003	**End Date:**	11/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Meyer, William V	**Contact Phone:**	216-433-5011
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Flight Program:	Foton		
Flight Assignment:	Foton M3 NOTE: Received NCE to 11/30/2008 per W. Meyer/GRC (3/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04GA45G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This work is part of an international collaboration between my own group at UC Santa Barbara funded by NASA, and that of professor Marzio Giglio at the University of Milan, funded by ESA/ASI. The flight hardware will be provided by ESA. Our aim is to carry out experimental studies of the fluctuations in liquids and liquid mixtures subjected to gradients. It is well known that macroscopic concentration and/or temperature disturbances relax to a homogeneous state via diffusion. Diffusion is caused by random molecular motion, and a common belief is that the fluctuations occur only on molecular length scales. This picture is wrong, and it is now understood that diffusion is associated with large amplitude, long wavelength fluctuations that restlessly accompany the decay to equilibrium. The fluctuations involved in a diffusion process are large, both in amplitude and spatial extent, and with the support from our first NASA grant, we were able to visualize them using the shadowgraph method. In order to provide a quantitative characterization of the fluctuations, we extended the theory of the shadowgraph, so as to provide quantitative results for the spatial power spectrum of the fluctuations, S(q). This technique allows us to measure the mean squared amplitude of the fluctuations as a function of wave vector q, i.e. S(q) . We have found that the shadowgraph method excels at very small wave vectors. The amplitude of the fluctuations diverges as q^-4 for intermediate wave vectors, so much of the interest will be at very small q, and we have thus focused our attention on the shadowgraph method. The divergence originates from the coupling of what are basically normal equilibrium velocity fluctuations with a macroscopic concentration and/or temperature gradient. Below a cutoff wave vector, the amplitude of the fluctuations is quenched to a constant value in the presence of gravity, but is predicted to continue diverging when gravity is removed. This divergence is predicted to be limited only by the finite size of the sample. Two of us (A. Vailati and M. Giglio) have reported on the observation of giant fluctuations during a free diffusion process occurring in ordinary fluids. These results provide strong evidence that giant concentration fluctuations are a genuinely universal phenomenon associated with diffusion caused by a gradient. The broad objectives of this work are to achieve a better understanding of the manner in which temperature and/or concentration gradients in fluid mixtures enhance spontaneous fluctuations. We also seek to elucidate the surprising role that gravity plays in suppressing such gradient driven fluctuations. Current predicitons are that removing gravity will result in extremely large long-wavelength fluctuations, limited in amplitude only by the size of the device containing the fluid. Further general objectives are to study the time dependence of such fluctuations as a function of the fluctuation wavelength. We intend to carry out such studies for both single-component fluids and for fluid mixtures. This experimental program impacts the way we understand diffusion processes and how we should model them when a temperature gradient or concentration gradient is present. To give one of many possible examples, when a protein crystal grows, it depletes the nutrient from the solute in which it is immersed. This depletion produces a concentration gradient, which in turn produces giant nonequilibrium concentration fluctuations. Thus, the current studies may impact our understanding of the growth process.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The main benefits are twofold. First, we have already developed new optical instrumentation making the equivalent of extremely low-angle light scattering possible, and we are gaining fundamental knowledge concerning the behavior of fluids and mixtures subjected to gradients. Our data will be used to test basic theories of fluid mechanics for non-equilibrium fluids.
Task Progress:	Since passing Critical Design Review in January, 2006, the contractors have constructed both the Qualification Model and the Flight Model for the experiment. Science Verification Testing was passed in March, 2007. ESA has now deemed the Flight Model accepatable and has taken delivery of the hardware. Mission Simulation Testing is scheduled for April 13-19, 2007 at ESTEC in Noordwijk, the Netherlands, and will be attended by the prime contractor, ESA officials, and the science teams from UC Santa Barbara and the University of Milan. This will be our one chance to prepare for the mission environment, and for the roles to be played by the various parties at the groundlink station ESRANGE in Kiruna, Sweden.. This test will also serve to provide the first ground-based data from the instrument, which is important because an important part of the science is to compare the amplitude and dynamical behavior of the fluctuation on the ground and in microgravity. Launch will be from Baikonur, Russia, September 14, 2007 and the flight duration will be 12 days. Efforts at UC Santa Barbara have focussed on developing a suitable method for the calibration of the device, which has turned out to pose non-trivial problems. We have now succeeded in calibrating the UCSB prototype instrument, after learning that we could suspend 2 micron diameter polystyrene latex spheres directly in isopropyl alcohol. (Attempts using 1 micron diameter melamine spheres, as previously planned, were not fully successful.) We have also developed the techniques necessary to detect particulate contamination of the sample, to remove such contamination, and to introduce the spheres without contaminating the sample. The first attempt at calibrating the Qualification Model failed because of problems with the integration of the electronics and software. Now that the Qualification Model is fully functional, and we have acquired the methods needed to carry out the calibration, we are hoping for a successful calibration. This will be done at the prime contractor's (RUAG) site in Zurich, Switzerland. In order to provide robust data analysis support for the mission, we have found it necessary to go beyond the in-house academic software we normally use which relies on the user to be expert in MatLab and/or Mathematica. For the mission, we have been writing true stand-alone programs for the Windows environment, with full graphical user interfaces. The required programs are: Two-dimensional Static Analysis, Display and Manipulation of Two-d Static Results, Two-d Dynamic Analysis, and Display and Manipulation of Two-d Dynamic Results. We now have functional beta level programs for the first two, and work is proceeding on the second two. We will use the first two during the Mission Simulation Test, which should reveal their value andj/or problems in time for us to fix them before the mission. Testing of the second two will be done using the ground-based data taken during the Mission Simulation Tests. The contractors have upgraded the Qualification Model to equal the Flight Model, and it will serve as a hands-on test bed during flight, should problems arrise or parameter tables require modification.
Bibliography Type:	Description: (Last Updated: 02/27/2013)

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